

Conjectures Related to *Lights Out*

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1 Definitions

Let *Lights Out* be a game played on a simple graph $G = (V, E)$ where clicking a vertex v toggles the on/off state of v and its neighbors. One wins the game by finding a sequence of clicks that turns off all the lights.

Definition 1. *The Most Clicks Problem (MCP) asks, for playing on some graph G , what is the most clicks needed to solve any solvable starting configuration, assuming that one solves each configuration in as few clicks as possible.*

That is, what is the most number of moves one needs to be able to solve any solvable initial configuration? This question is akin to looking for “God’s Number” on a Rubik’s Cube.

Definition 2. *Let $d(n) = \dim(\ker(A + I))$ over the field $\mathbb{Z}_2$ where A is the adjacency matrix of G an $n \times n$ grid.*

You may also hear $d(n)$ referred to as the *nullity* of the $n \times n$ grid. The right-hand side of the equation applies for any graph, not just grids. Elements of the null space correspond to sets of clicks that have no net effect on the states of the vertices when applied. These elements also correspond exactly to subsets of V such that all vertices have an even number of elements from the subset in their closed neighborhood (themselves and their neighbors). Thus, these elements of the null space may be referred to as *null patterns*, *quiet patterns*, or *even dominating sets*.

Definition 3. *Let $F_n(x)$ be the polynomial in the ring $\mathbb{Z}_2[x]$ such that*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xF_{n-1}(x) + F_{n-2}(x) & n > 1 \end{cases}.$$

These polynomials become useful in calculating $d(n)$. It is known that

$$d(n) = \deg \gcd(F_{n+1}(x), F_{n+1}(x+1)).$$

2 Conjectures

The mixed-prime formulas formerly listed here are now theorems. The finite-support truncation theorem in `tex/multiprime_stabilization/multiprime_stabilization.tex` reduces every fixed odd prime

support to finitely many exponent vectors, and exact computations settle the previously listed two-prime formulas. The same theorem proves that $d(a^k - 1)$ is eventually constant for every odd positive integer a , so the former eventual-constancy conjecture has also been removed.

We also have conjectures about $d(p^k - 1)$ for p a prime.

Conjecture 1. *Let p be a Wieferich prime. Then for every $k \geq 1$,*

$$d(p^k - 1) = d(p - 1).$$

The formula is already proved for every non-Wieferich prime. More generally, if

$$c_p = v_p(2^{\text{ord}_p(2)} - 1),$$

then prime-power stabilization proves

$$d(p^k - 1) = d(p^{c_p} - 1) \quad (k \geq c_p).$$

The two known Wieferich primes, 1093 and 3511, both have $c_p = 2$, and exact polynomial GCD computations give

$$d(p^2 - 1) = d(p - 1) = 0$$

for both. Thus the conjecture holds for every currently known Wieferich prime; any counterexample requires a presently unknown one.

Proposition 1. *Let $p > 3$ be prime and define*

$$R_p = \{\zeta + \zeta^{-1} : \zeta \in \overline{\mathbb{F}_2}, \text{ord}(\zeta) = p\}.$$

Then

$$d(p - 1) = 2|R_p \cap (R_p + 1)|.$$

In particular, $d(p - 1) \neq 0$ if and only if there exist nontrivial p th roots of unity t, s satisfying

$$1 + t + t^{-1} + s + s^{-1} = 0.$$

Equivalently, p is a prime mad number in the terminology of Button Madness.

Blokhuis proved the necessary condition

$$d(p - 1) \neq 0 \implies \rho_2(p) \leq \sqrt{p}, \quad \rho_2(p) = \min\{r > 0 : 2^r \equiv \pm 1 \pmod{p}\}.$$

The converse is false: $\rho_2(241) = 12 < \sqrt{241}$, but $d(240) = 0$. On the positive side, every Mersenne prime greater than 7 and every Fermat prime greater than 3 has nonzero $d(p - 1)$.

Problem 1. *Give a purely arithmetic characterization of the primes p for which*

$$R_p \cap (R_p + 1) \neq \emptyset.$$

Conjecture 2. *There are infinitely many primes p for which $d(p - 1) \neq 0$.*

We know that $d(n) = 26$ is the smallest even impossible value for d . We also know that if a is an impossible even value of d , then $2a + 2$ is also impossible. We would like to better classify and understand impossible values. Let I be the set of even values that are never values of $d(n)$. Call an element of I a *seed* if it is not obtained from a smaller impossible value by the recurrence $a \mapsto 2a + 2$.

Conjecture 3. *The following are true:*

1. $34 \in I$ (implies $70, 142, 286, \dots \in I$)

2. *I has infinitely many seed values.*

Our method for solving the MCP for nullity 2 grids involved getting an exact solution to an integer linear program by noticing all constraints could be tightly satisfied at once. We proved that every nullity 2 grid has size $6k - 1$ and that its MCP value is $26k^2 - 12k + 1$. However, this does not seem to work for higher nullities. We suspect that the answer to the MCP is still a parabola though. Thus, we can calculate the answer to the MCP for 3 boards of the same nullity in the same family (i.e. size is same mod some number) and derive the parabola.

Conjecture 4. *Regarding the answer to the MCP for an $n \times n$ grid,*

1. *If $n = 10k - 6$ and $d(n) = 4$, then the answer to the MCP is $17k^2 - 10k$.*
2. *All nullity 4 grids are of size $10k - 6$*
3. *If $n = 24k - 13$ and $d(n) = 6$, then the answer to the MCP is $88k^2 - 24k + 1$.*
4. *All nullity 6 grids are of size $24k - 13$.*
5. *If $n = 40k - 31$ and $d(n) = 8$, then the answer to the MCP is $60k^2 - 20k - 3$.*
6. *If $n = 34k - 18$ and $d(n) = 8$, then the answer to the MCP is $161k^2 - 34k - 3$.*
7. *If $n = 60k - 31$ and $d(n) = 10$ then the answer to the MCP is $506k^2 - 60k - 3$.*
8. *All nullity 10 grids are of size $60k - 31$.*
9. *If $n = 340k - 256$ and $d(n) = 12$, then the answer to the MCP is $3761k^2 - 169k - 13$.*
10. *All nullity 12 grids are of size $340k - 256$.*
11. *...*

Notice that there are different “families” associated with each nullity. These families correspond to a certain size of board. Notice that one nullity might have multiple families. It seems that the smallest board in each family appears in OEIS sequence A118142. We suspect that the parabolas giving the answer to the MCP for a given family always have integer coefficients.

We also have a conjecture about the maximum values of $d(n)$.

Conjecture 5.

$$d(n) \leq \frac{4(n+1)}{5},$$

with equality if and only if $n = 5 \cdot 2^k - 1$.

The family $n = 5 \cdot 2^k - 1$ is already known to attain equality. The unresolved parts are the global upper bound and the exclusion of any other equality cases. As far as we know, the only other proof on bounds of $d(n)$ known to date is that $d(4) = 4$ is the only case where $d(n) = n$.